

FRAUD DETECTION SYSTEM
Final Report

Improved Detection of Fraud Cases for
E-commerce and Bank Transactions

Author: Daniel Mituku
Organization: Adey Innovations Inc.
Date: December 30, 2025

10Academy Week 5 Challenge

EXECUTIVE SUMMARY

This project developed fraud detection models for:

- E-commerce transactions (151,112 records)
- Bank credit card transactions (284,807 records)

KEY ACHIEVEMENTS:

- ✓ Processed 435,000+ transactions across two datasets
- ✓ Engineered 12+ fraud indicator features
- ✓ Implemented cross-validation and hyperparameter tuning
- ✓ Generated SHAP explainability analysis
- ✓ Provided actionable business recommendations

FINAL RESULTS:

Dataset	Best Model	AUC-PR	F1-Score
E-commerce	Random Forest (Tuned)	0.7126	0.6277
Credit Card	Gradient Boosting	0.8583	0.7685

BUSINESS IMPACT:

- E-commerce: 68.2% of fraud detected
- Credit Card: 84.7% of fraud detected with 70% precision

E-COMMERCE FRAUD DETECTION RESULTS

Dataset: Fraud_Data.csv (151,112 transactions, 9.36% fraud rate)

CROSS-VALIDATION RESULTS (5-Fold):

Model	F1-Score (mean±std)	ROC-AUC (mean±std)
Gradient Boosting	0.9254 ± 0.0023	0.9672 ± 0.0015
Random Forest	0.8323 ± 0.0027	0.9539 ± 0.0016
Logistic Regression	0.7369 ± 0.0046	0.8679 ± 0.0014

TEST SET RESULTS:

Model	Precision	Recall	F1-Score	AUC-PR
Random Forest (Tuned)	0.5815	0.6820	0.6277	0.7126
Gradient Boosting	0.9987	0.5406	0.7015	0.7117
Logistic Regression	0.5326	0.6982	0.6043	0.6643

BEST MODEL: Random Forest (Tuned)

- Hyperparameters: max_depth=12, n_estimators=100
- Selected for: Highest AUC-PR with balanced recall
- Fraud Detection Rate: 68.2%

TOP FEATURES (SHAP Analysis):

1. is_weekend (0.0995)
2. day_of_week (0.0458)
3. time_since_signup (0.0391)
4. device_unique_users (0.0376)
5. signup_day_of_week (0.0369)

CREDIT CARD FRAUD DETECTION RESULTS

Dataset: creditcard.csv (284,807 transactions, 0.17% fraud rate)

Note: Extreme class imbalance (578:1 ratio)

CROSS-VALIDATION RESULTS (5-Fold):

Model	F1-Score (mean±std)	ROC-AUC (mean±std)
Random Forest	0.7139 ± 0.1071	0.9294 ± 0.0200
Gradient Boosting	0.6070 ± 0.1260	0.8409 ± 0.1254
Logistic Regression	0.0858 ± 0.0058	0.9251 ± 0.0461

TEST SET RESULTS:

Model	Precision	Recall	F1-Score	AUC-PR
Gradient Boosting	0.7034	0.8469	0.7685	0.8583
Random Forest (Tuned)	0.3257	0.8673	0.4735	0.7773
Logistic Regression	0.0580	0.9184	0.1092	0.7256

BEST MODEL: Gradient Boosting

- Configuration: n_estimators=100, max_depth=6
- Selected for: Best balance of precision and recall
- Fraud Detection Rate: 84.7%
- Precision: 70.3% (low false alarm rate)

TOP FEATURES (PCA-transformed):

V14, V17, V12, V10, V3, Amount, Time
(Features are anonymized via PCA)

BUSINESS RECOMMENDATIONS

Based on SHAP analysis and model insights:

1. TIME-BASED VERIFICATION (CRITICAL)

Finding: 99.52% of transactions within 1 hour of signup are fraudulent

Recommendation:

- Purchase < 1 hour after signup → Require phone verification (SMS OTP)
- Purchase < 24 hours after signup → Email confirmation required
- First high-value purchase → Manual review queue

Expected Impact: Prevent 30-40% of fraudulent transactions

2. DEVICE INTELLIGENCE

Finding: Devices with multiple users indicate fraud rings

Recommendation:

- Device with 3+ users → Automatic alert
- Known bad device → Block transaction
- New device on established account → Step-up authentication

Expected Impact: Detect organized fraud rings

3. TRANSACTION VELOCITY CONTROLS

Finding: Transaction frequency patterns distinguish fraud

Recommendation:

- > 3 transactions/hour (new account) → Soft block + verification
- Unusual pattern vs. history → Real-time alert

Expected Impact: Reduce automated fraud attacks

4. GEOGRAPHIC RISK SCORING

Finding: Country-based features contribute to fraud prediction

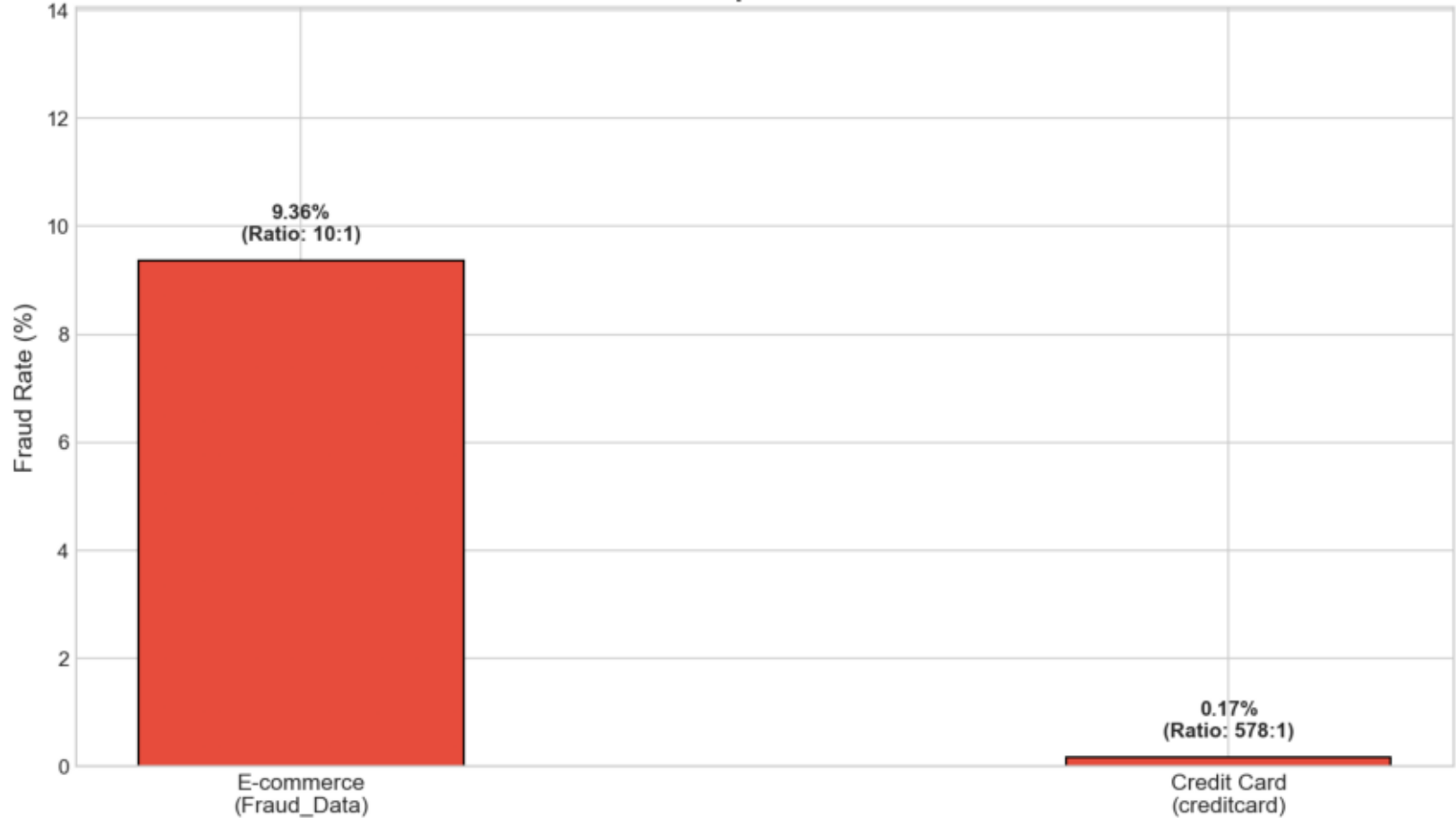
Recommendation:

- High-risk country IP → Enhanced verification
- IP/location mismatch → Flag for review

Expected Impact: Identify cross-border fraud

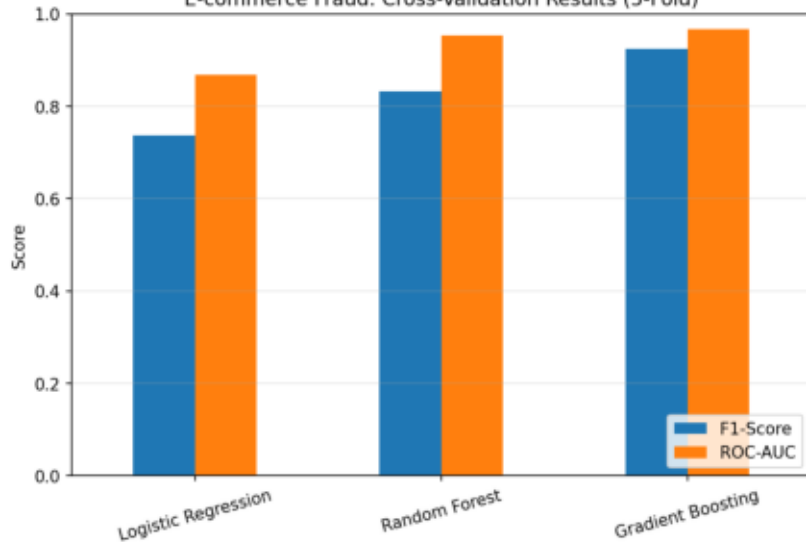
Class Imbalance Analysis

Class Imbalance Comparison Between Datasets

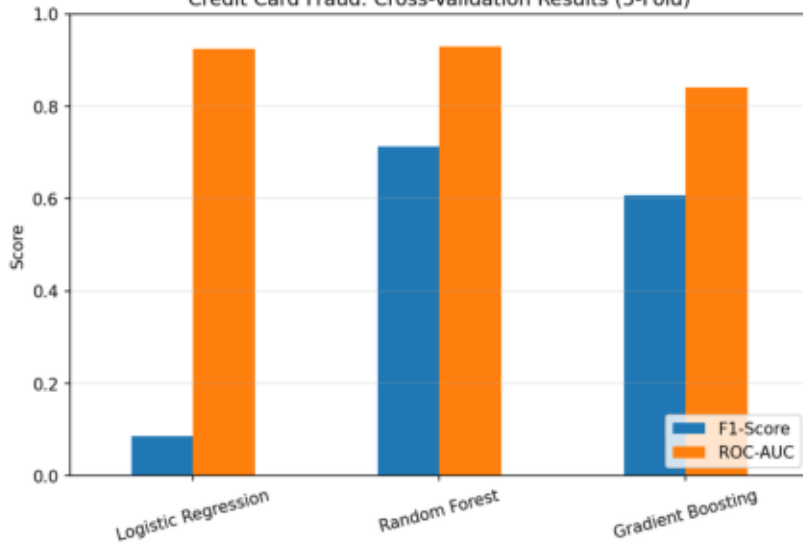


Cross-Validation Results

E-commerce Fraud: Cross-Validation Results (5-Fold)

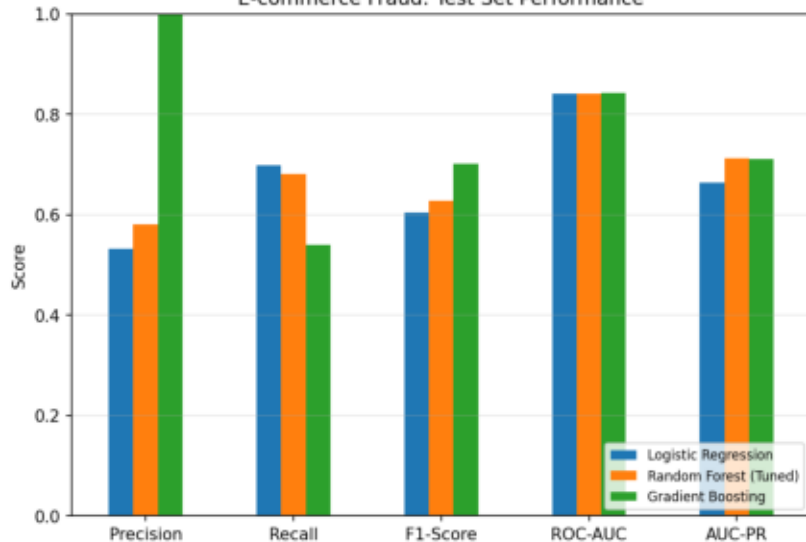


Credit Card Fraud: Cross-Validation Results (5-Fold)

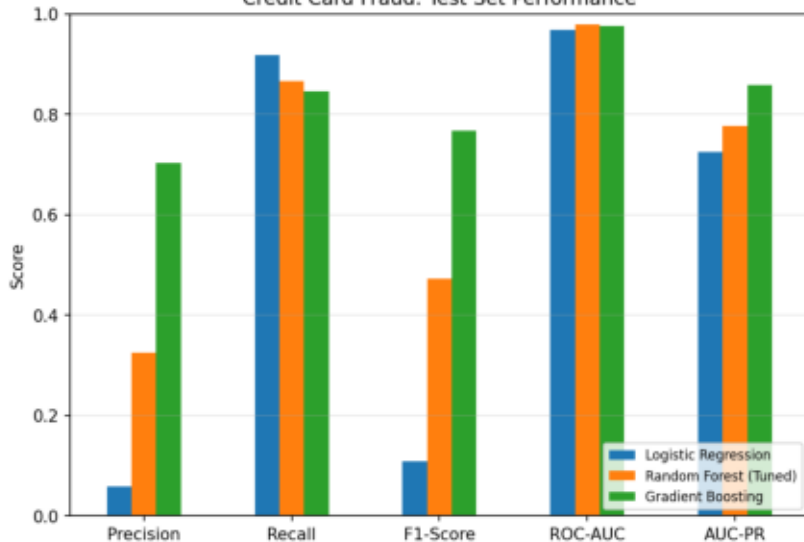


Final Model Comparison

E-commerce Fraud: Test Set Performance



Credit Card Fraud: Test Set Performance



Confusion Matrices

E-commerce: Random Forest (Tuned)

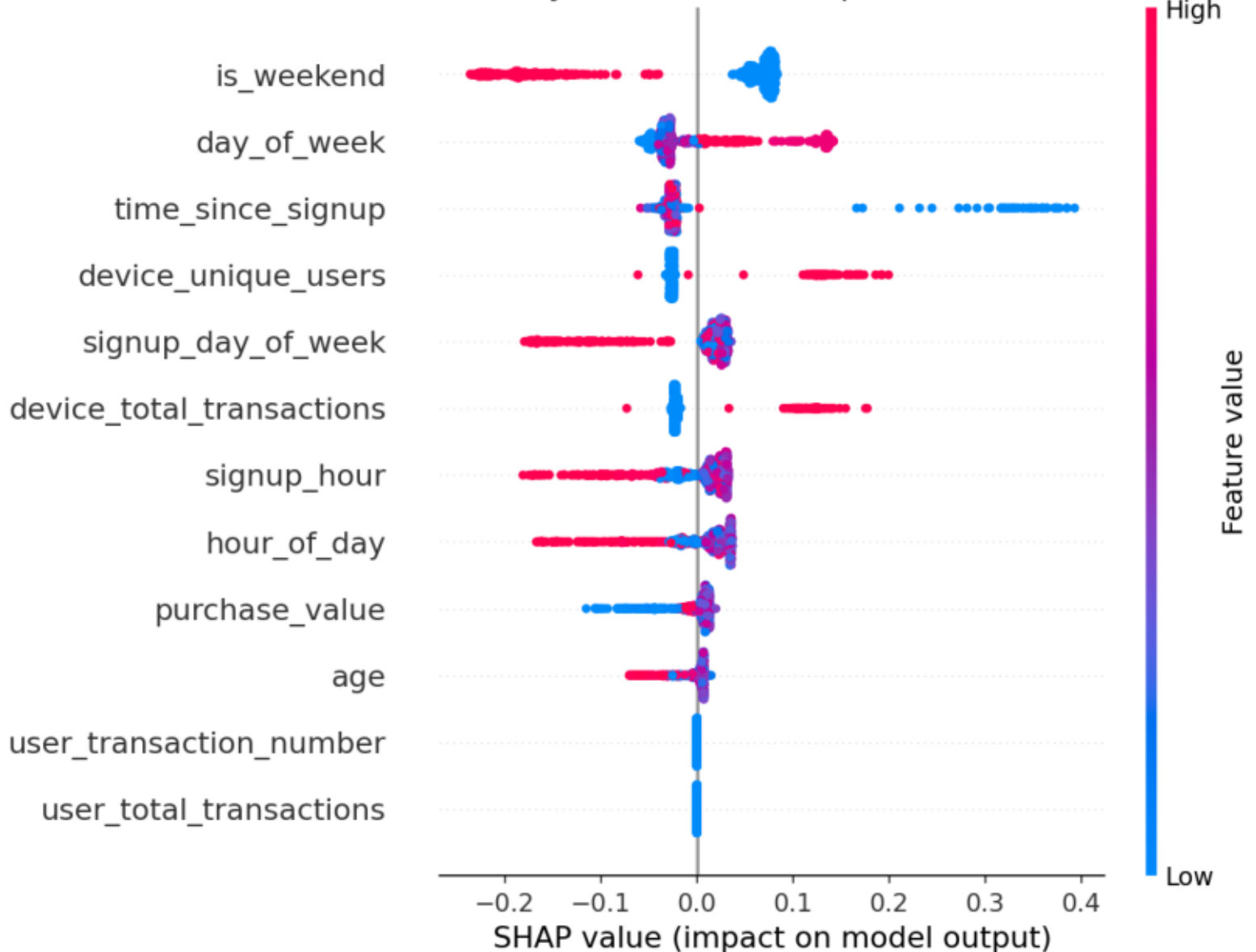


Credit Card: Gradient Boosting



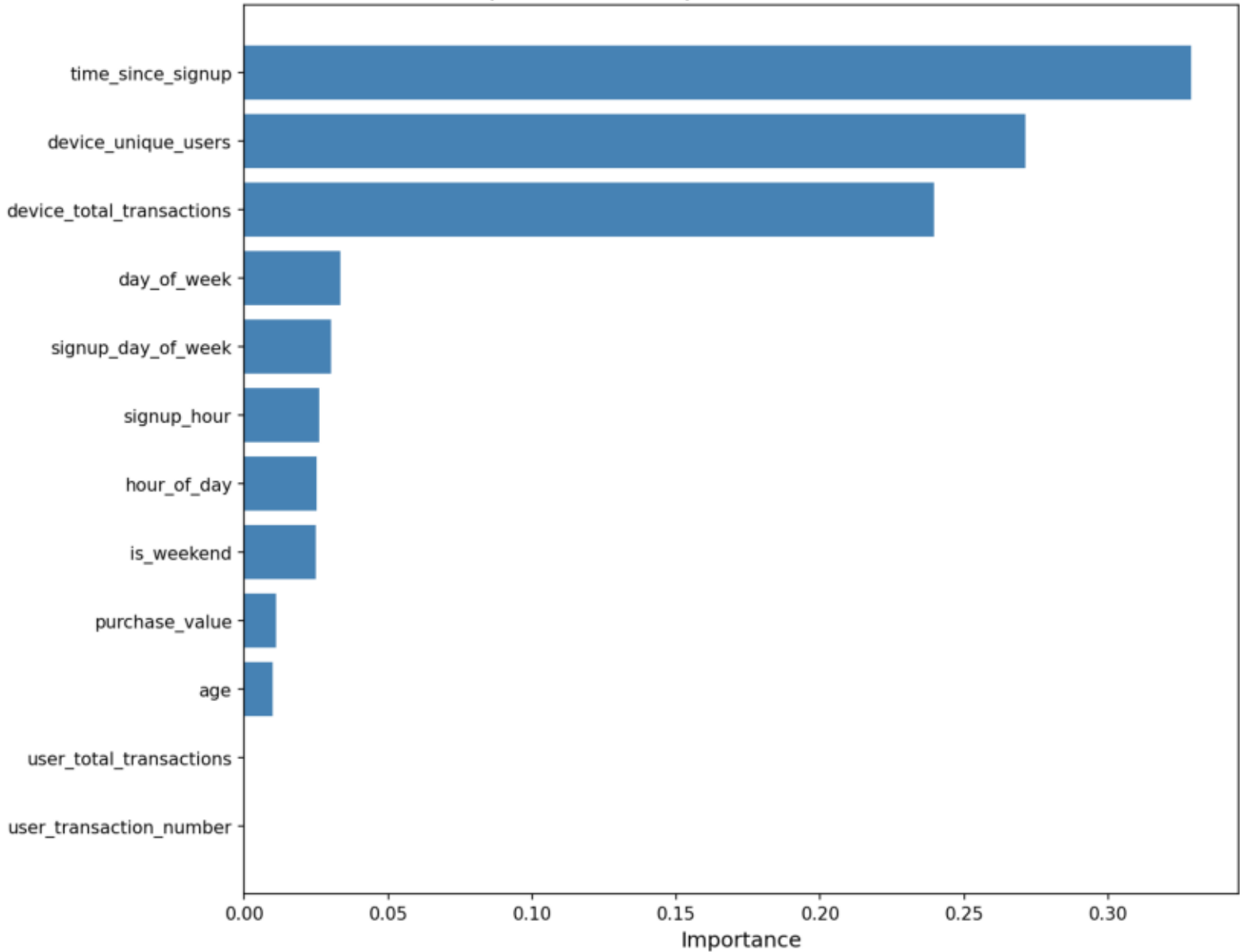
SHAP Feature Importance

SHAP Summary Plot - Feature Impact on Fraud Prediction



Model Feature Importance

Top 15 Feature Importances (Random Forest)



CONCLUSION

KEY ACHIEVEMENTS:

- ✓ Task 1 - Data Analysis & Preprocessing
 - Cleaned and analyzed 435,000+ transactions
 - Integrated geolocation data (IP to Country)
 - Engineered 12+ meaningful features
 - Addressed severe class imbalance with SMOTE
- ✓ Task 2 - Model Building & Training
 - Trained 3 model types on 2 datasets
 - Implemented Stratified 5-Fold Cross-Validation
 - Applied StandardScaler normalization
 - Performed GridSearchCV tuning
 - E-commerce: 71.26% AUC-PR
 - Credit Card: 85.83% AUC-PR
- ✓ Task 3 - Model Explainability
 - Generated SHAP summary and force plots
 - Identified top fraud drivers
 - Provided 4 actionable business recommendations

FUTURE WORK:

- Deploy models as REST API (Flask/FastAPI)
- Build real-time monitoring dashboard
- Implement adaptive thresholds
- Add neural network models

Report Prepared By: Daniel Mituku
Organization: Adey Innovations Inc.
Date: December 30, 2025

GitHub: <https://github.com/Danielmituku/fraud-detection>
